CHAPTER 3

SAMPLING TECHNIQUES

CHAPTER ESSENCE AT A GLANCE:
Master Census versus Sample inquiry, probability sampling designs (SRS, Stratified, Systematic, Cluster), non-probability designs, and sampling vs non-sampling errors.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Census vs Sample Investigation:** Census involves complete enumeration of every unit in the population, offering high accuracy at massive cost; Sampling inspects a representative subset, saving time and resources.

• **Simple Random Sampling (SRS):** Every unit in the population has a known, equal, and independent chance of selection; executed via lottery method or random number tables (with or without replacement).

• **Stratified and Systematic Sampling:** Stratified sampling partitions heterogeneous populations into homogeneous strata, sampling within each; Systematic sampling selects every k-th item after a random start.

• **Sampling vs Non-Sampling Errors:** Sampling errors arise purely from observing a sample rather than the full population (decreases as sample size expands); Non-sampling errors arise from measurement defects, non-response, and recording errors (increases with population size).

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$\text{Sampling Interval} \Longleftrightarrow k = \frac{N}{n} \rightarrow \text{Select Units } \{r, r+k, r+2k, \ldots\}$$

Systematic Random Sampling Interval Formula

$$\text{Stratified Proportional Allocation} \Longleftrightarrow n_i = n \times \frac{N_i}{N}$$

Sub-Sample Size Proportional to Stratum Population

$$\text{Total Survey Error} \equiv \text{Sampling Error} + \text{Non-Sampling Error}$$

Dual Composition of Empirical Statistical Error

$$\text{Sampling Error Magnitude} \propto \frac{1}{\sqrt{n}} \rightarrow \lim_{n \to N} (\text{Sampling Error}) = 0$$

Inverse Relationship Between Sample Size and Sampling Error

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Sampling error vanishes completely in a complete census, but non-sampling error can increase significantly due to massive administrative scale.

• **Alert:** In Stratified Sampling, strata must be internally homogeneous and externally heterogeneous; in Cluster Sampling, clusters must be internally heterogeneous and externally homogeneous.

• **Alert:** Convenience sampling and quota sampling are non-probability methods; results cannot be evaluated for statistical sampling error.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between Census Method and Sampling Method with two points of difference and state when Census is essential. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Differences:** (1) Census studies each unit of the universe; Sample studies only a representative portion. (2) Census is time-consuming and expensive; Sample is fast and economical. **[1.5 Marks]**

• **When Census is Essential:** When the universe is small, units are highly heterogeneous, or when extreme precision is legally mandated (e.g. National Population Census, blood testing of a patient requires sampling because census destroys universe). **[1.5 Marks]**

2. Explain Simple Random Sampling and Stratified Random Sampling. Give one practical situation suitable for each. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Simple Random Sampling:** A method where every individual unit of the population has an equal, known, and independent probability of being included in the sample. Suitable for homogeneous populations (e.g. selecting scholarship recipients among identical high performers). **[2 Marks]**

• **Stratified Random Sampling:** Population is divided into non-overlapping groups called strata such that units within a stratum are homogeneous, and samples are drawn from each stratum. Suitable for heterogeneous populations (e.g. studying household expenditure across rural, urban, and semi-urban income brackets). **[2 Marks]**

3. Explain the Systematic Sampling procedure. How is the sampling interval calculated? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Procedure:** Units in the population are numerically arranged from 1 to N. A random number r between 1 and k is chosen as the starting point, and every k-th unit thereafter is selected into the sample. **[1.5 Marks]**

• **Formula:** Sampling interval k = N divided by n, where N = Total population size and n = Desired sample size. The resulting sample comprises units: r, r + k, r + 2k, ..., r + (n-1)k. **[1.5 Marks]**

4. Distinguish between Sampling Errors and Non-Sampling Errors. Which error can occur in a Census? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Sampling Errors:** Errors arising solely because only a sample of units is observed instead of the entire population; decreases as sample size n increases. **[1 Mark]**

• **Non-Sampling Errors:** Errors arising from data collection defects, non-response, ambiguous questions, faulty measurement instruments, and clerical tabulation errors. **[1 Mark]**

• **Error in Census:** In a complete Census, sampling errors are zero, but non-sampling errors are present and often substantial due to the massive volume of operations. **[1 Mark]**

5. What is the difference between Simple Random Sampling With Replacement (SRSWR) and Without Replacement (SRSWOR)? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **SRSWR:** A selected unit is recorded and returned to the population before the next draw; probability of selection remains constant at 1 divided by N. **[1 Mark]**

• **SRSWOR:** A selected unit is not returned; the population size reduces by one with each draw, making selection probabilities 1 divided by N, 1 divided by (N-1), and so on. **[1 Mark]**

6. State two examples of non-probability sampling methods and explain why they are unsuited for hypothesis testing. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Examples:** Judgement Sampling (purposive) and Quota Sampling. **[1 Mark]**

• **Reason:** Selection probabilities are unknown and subjective; hence, standard errors and sampling error margins cannot be calculated mathematically. **[1 Mark]**